AP - 18.2 - 14 16.12.2026

Lymphoid & Myeloid stem cells do not immediately divide & differentiate into mature formed element's

These cells have suffix blast as being the precursor cells

e.g → megakaryoblasts are the precursors of megakaryocytes, & proerythroblasts become reticulocytes → which eject their nucleus & most other organelles before maturing mutacing into erythrocytes.

Hemopoietic Growth Factors →

Development from stem cells to precursor cells to mature cells is again initiated by hemopoietic growth factors

- Erythropoietin (EPO) → is a glycoprotein hormone secreted by the interstitial fibroblast cells of the kidneys in response to low O_2 levels. It prompts the production of erythrocytes.

Some athletes use synthetic EPO as a performance-enhancing drug (doping) to increase RBC counts & subsequently increase O_2 delivery to tissues throughout the body EPO → is used to treat anemia, some types of cancer.

Thrombopoietin → another glycoprotein hormone is produced by the liver & kidneys

It triggers the development of mega-karyocytes & platelets